

Product and Quotient Laws for Integer Exponents

Expanded-form models, the product law, and the quotient law

The model behind both laws

- An exponent counts repeated factors. Writing a power out in expanded form is always allowed, and it is what makes each law obvious.
 - **Multiplying** puts two groups of factors end to end, so the counts add: $x^m \cdot x^n = x^{m+n}$.
 - **Dividing** cancels matching factors top and bottom, so the counts subtract: $\frac{x^m}{x^n} = x^{m-n}$.
 - Leave answers as a single power of the given base, with no negative exponent in the final line unless a problem asks for one.
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1. Expanded-form model. Write $2^3 \cdot 2^4$ out as a string of factors of 2, then count them.

(a) Written as a single power of 2, the exponent is _____.

(b) Its value is _____.

2. Simplify to a single power: $x^5 \cdot x^3$.

Answer: _____

3. Simplify to a single power: $\frac{y^9}{y^4}$.

Answer: _____

4. Expanded-form model. Write $\frac{5^6}{5^4}$ with the factors of 5 shown top and bottom, then cancel the matching pairs.

(a) Written as a single power of 5, the exponent is _____.

(b) Its value is _____.

5. Simplify to a single power: $m^7 \cdot m^{-3}$.

Answer: _____

6. Simplify, and write the answer with a positive exponent: $\frac{t^4}{t^{10}}$.

Answer: _____

7. Simplify: $(2x^3)(5x^4)$.
Handle the number factors and the powers separately.

Answer: _____

8. Simplify: $\frac{12n^8}{4n^5}$.
Handle the number factors and the powers separately.

Answer: _____

9. Write $<$, $>$, or $=$ between these two quantities, using the laws rather than working out both values the long way.

$$2^3 \cdot 2^5 \quad \underline{\hspace{1cm}} \quad \frac{2^{10}}{2^2}$$

10. Challenge. Jonah writes $x^4 \cdot x^3 = x^{12}$.

- (a) Give the correct simplification. Answer:
- (b) Explain why the exponents are added and not multiplied here. Use the expanded-form model in your reason.